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ABSTRACT

Text extraction from images, commonly referred to as Optical Character Recognition (OCR), is essential for converting visual information into machine-readable text. While OCR technology has advanced significantly, it continues to face challenges when dealing with handwritten text, varying fonts, complex backgrounds, and poor lighting conditions. This report presents a deep learning-based approach to enhance the accuracy of text extraction by utilizing Convolutional Neural Networks (CNN) for feature extraction and classification, integrated with an end-to-end OCR pipeline.

The proposed model employs a CNN architecture to detect and classify characters from both printed and handwritten text images. The CNN automatically learns spatial features such as edges, curves, and textures, enabling robust recognition across a wide range of character styles. The model is trained and validated on a comprehensive dataset comprising the EMNIST (Extended Modified National Institute of Standards and Technology) dataset and a custom-built dataset of real-world handwritten and printed images. The data includes 39 categories: alphabets, digits, and select special characters.

Experimental results demonstrate that the model achieves high accuracy in recognizing characters, even under noisy or low-quality conditions. Performance metrics such as accuracy (92.36%), precision (93.06%), recall (92.52%), and F1-score (92.64%) confirm the reliability of the CNN-based system in diverse testing environments. While the model performs well across various text formats, future enhancements could focus on multilingual support, sequence-based word recognition, and real-time deployment on mobile platforms.

This project emphasizes the potential of combining deep learning with OCR techniques to develop more accurate, adaptable, and scalable text recognition systems suitable for real-world applications such as document digitization, automated data entry, and assistive reading technologies.

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Chapter 1

INTRODUCTION

Text extraction from images, or Optical Character Recognition (OCR), involves detecting and converting text from various visual sources such as scanned documents, street signs, forms, and handwritten notes into machine-readable format. This process is widely used in areas like document digitization, intelligent form processing, and real-time translation from printed materials. However, accurate text recognition remains ¹³ a challenging task due to the presence of noise, distortions, complex backgrounds, and varied text orientations in real-world images.

1.1 Background

Despite significant advances in OCR, traditional techniques often struggle with complex scenarios involving handwritten characters, inconsistent fonts, skewed text, and background clutter. These limitations reduce their reliability in practical applications. ²⁷ Conventional machine learning models such as Support Vector Machines (SVM) ²⁷ were previously used to segment and classify text regions, but they rely heavily on manual feature extraction and are less effective when dealing with noisy or distorted images.

To address these challenges, this project adopts a deep learning approach using ³¹ Convolutional Neural Networks (CNNs), which have ²² shown great success in image classification tasks. Unlike traditional methods, CNNs ²² automatically learn spatial hierarchies of features from raw pixel data, making them more robust to variations in writing style, font, and image quality. CNNs also excel in recognizing patterns in complex visual inputs, allowing them to handle diverse text appearances more effectively.

This project focuses on building a deep learning-based OCR framework that integrates CNNs for feature extraction and character classification. The system is

trained using the EMNIST dataset, which contains handwritten characters, as well as a custom dataset of printed and handwritten text images captured under real-world conditions. With proper preprocessing and data handling, the CNN model is capable of recognizing text with high accuracy across varied image types. The approach not only simplifies the recognition pipeline but also enhances performance in noisy and uncontrolled environments.

1.2 Problem Statement

Extracting text from images becomes especially difficult when dealing with handwritten content, low-resolution images, uneven lighting, and background noise. Traditional OCR systems often fail to generalize well in such conditions due to their dependence on manually engineered features and rule-based processing. These systems struggle with recognizing distorted, overlapping, or multi-oriented text, making them unsuitable for real-world deployment.

To overcome these issues, this project implements a Convolutional Neural Network (CNN)-based model capable of learning and identifying text patterns directly from pixel-level data. By using deep learning, the system aims to improve character recognition accuracy and robustness, even in images with significant variations. The goal is to move beyond the limitations of conventional OCR systems and build a more adaptive and scalable solution.

1.3 Objectives

The primary objective of this project is to develop a robust and efficient deep learning-based OCR system using CNN for accurate text extraction from images. The system is designed to address key challenges such as inconsistent lighting, diverse fonts, distorted characters, and handwritten content. The following are the specific objectives of the project:

Automatic Feature Learning: Enable the model to automatically detect important visual patterns from input images using convolutional layers, without manual feature

engineering.

Improved Text Recognition: Accurately recognize and classify characters from both printed and handwritten images, using a unified CNN architecture.

Robustness Across Conditions: Ensure stable performance across images with noise, varying backgrounds, and different text orientations.

Practical Applications: Design the system for real-world use cases, including document scanning, automatic form reading, and mobile-based image-to-text translation.

Real World Use Cases: The objective is also to develop a system that has real life application, for example, in capturing text from documents, reading through forms automatically, and translating text on the image in real time.

Chapter 2

LITERATURE SURVEY

¹ As discussed, earlier text recognition from images is still an active research in the field of pattern recognition. To address the issues related to text recognition many researchers have proposed different technologies, each approach or technology tries to address the issues in different way. We have presented a detailed survey of approaches proposed to handle the issues related to text recognition.

Text detection and recognition in complex and cluttered environments has gained significant attention due to its applications in document analysis, scene text extraction, and image processing. Various models and techniques ²⁹ have been proposed to address the challenges posed by diverse text styles, orientations, and backgrounds. This section reviews several notable approaches in the field, highlighting both their strengths and limitations, particularly in handling complex scenes.

Optical Character Recognition (OCR) has advanced from simple document scanning to complex text recognition in natural scenes. Yuming He [1] highlighted its role in converting images to machine-readable text, addressing challenges such as handwriting styles and complex backgrounds. Traditional methods like SVM were effective for printed text but struggled with scene variability. Modern deep learning-based OCR systems significantly enhance functionality.

Deepa et al. [2] explored the application of Support ³⁰ Vector Machines (SVMs) and Convolutional Neural Networks (CNNs) in image classification. SVMs were highlighted for their ability ⁶ to handle high-dimensional data efficiently, leveraging margin maximization to achieve robust generalization. The flexibility of kernel functions, including linear kernels, enables SVMs to perform both linear and non-linear classification, making them valuable in domains like medical imaging and

document classification. CNNs, on the other hand, excel in hierarchical feature extraction through convolution and pooling layers, automating filter optimization and capturing abstract features, which significantly enhances performance in tasks like object recognition.

Ghai et al. [3] emphasized the role of K-means clustering in text detection, highlighting its simplicity, efficiency, and adaptability. As an unsupervised learning algorithm, K-means effectively handles heterogeneous datasets, making it suitable for detecting text with varying fonts, sizes, orientations, and backgrounds. The study applied K-means to segment high-frequency wavelet features extracted from decomposed image. By iteratively minimizing intra-cluster variance and maximizing inter-cluster separation, the algorithm optimized cluster centers using statistical measures like mean and standard deviation of wavelet coefficients. This approach proved effective in distinguishing text from non-text regions in complex image datasets.

Tang et al. [4] introduced Cascaded Convolutional Neural Networks (CNNs) for text detection and segmentation, emphasizing their hierarchical processing capabilities. This architecture refines outputs progressively, enabling effective extraction of local features like edges and textures, as well as global context. By structuring tasks into extraction, refinement, and classification phases, cascaded CNNs enhance precision while minimizing false positives. The multi-layer refinement strategy resolves ambiguities and consolidates overlapping text regions, ensuring high accuracy and reliability even in complex scenes with various text styles, orientations, and sizes. This systematic approach makes Cascaded CNNs highly effective for robust text detection tasks.

Rong et al. [5] proposed a novel approach for text segmentation in cluttered scenes by integrating visual and linguistic features through referring expressions. The model utilizes both visual data and natural language descriptions, enabling effective segmentation, and was evaluated on the COCO-CharRef dataset. This method

significantly enhances contextual text segmentation by combining multimodal inputs, improving accuracy in complex scenarios. However, the reliance on annotated datasets limits the model's adaptability to new environments, highlighting a need for more generalized solutions.

Yousef et al. ¹⁶ [6] proposed a CNN-based model for handwriting recognition that demonstrates high efficiency in handling both short and long lines of text across various handwriting styles and sizes. The model effectively recognizes unconstrained text with minimal data, making it versatile for diverse applications. However, its performance declines when applied to paragraphs or multiple lines of text without additional segmentation algorithms, highlighting a limitation in processing extensive textual content.

Surana et al. [7] reviewed several methodologies for character recognition, emphasizing their evolution and effectiveness. Back Propagation Networks (BPN) were among the earliest techniques, leveraging the Radon Transform combined with Back Propagation Neural Networks for image-to-binary conversion. This method segmented images into sub-images, converting individual characters into binary format, which proved instrumental for scanned text recognition. Artificial Neural Networks (ANNs) advanced these efforts by introducing a three-layer architecture optimized using the Scaled Conjugate Gradient method. This approach surpassed traditional backpropagation techniques, achieving a recognition accuracy of 95%, thereby establishing ANNs as a reliable model for text classification.

Chapter 3

METHODOLOGY

Text extraction from images, also known as Optical Character Recognition (OCR), involves detecting and extracting text from a variety of image sources such as street signs, posters, documents, or handwritten notes. This process is a critical component of many applications, including document digitization, automated form reading, and real-time text translation from street signs or product labels.

3.1 Data Collection and Preparation

⁴⁴ The model is trained and tested on the ¹⁵ EMNIST (Extended Modified National Institute of Standards and Technology) dataset, which contain over 857,000 images of handwritten English alphabets and digits. The EMNIST dataset is selected due to its large sample size and wide variety of handwriting styles, making it suitable for training a deep learning model with high generalization capabilities. In this project, a twofold approach was employed to prepare a comprehensive dataset.

² To prepare the data for training, several preprocessing techniques were applied. All images were transformed into 32×32 pixel black-and-white images to ensure uniformity across the dataset. Grayscale images were binarized using thresholding techniques to enhance character contrast and remove background noise. In terms of class categorization, the dataset was carefully structured to avoid misclassification. A total of 39 categories were created, including 26 alphabet characters (with uppercase and lowercase letters combined into single classes), 9 numeric digits (1 through 9), and 4 commonly used special characters (@, #, \$, &). Notably, digit '0' was merged with the character 'O' due to their high visual similarity in handwritten form, which often leads to misclassification in OCR systems.

In addition to the EMNIST dataset, a custom dataset was developed to evaluate the

model's robustness in real-world scenarios. This dataset consists of images captured manually using a mobile device and includes a mixture of handwritten and printed text. These real-world images were taken under varying lighting conditions and backgrounds, simulating challenges typically encountered in practical OCR applications. The custom images were also resized to 32×32 pixels and processed using noise filtering, contrast enhancement, and normalization techniques to align with the training data format.

The final dataset represents a total of 39 distinct classes and includes both structured data from EMNIST and unstructured, real-world data from user-captured images. This combination ensures that the model is exposed to a wide range of character styles, fonts, and noise conditions, enhancing its ability to generalize well in both controlled and uncontrolled environments.

3.2 Workflow

The workflow for the text extraction system is designed as a step-by-step process that converts raw images into recognized text using a Convolutional Neural Network (CNN). This structured pipeline ensures consistent performance and adaptability in various environments, including both scanned and real-world captured images. Below is a brief description of each step in the sequence as shown in Fig 3.1.

- **Data Loading:** This is the initial step where images are collected from both the EMNIST dataset and a custom dataset created using mobile phone photography. These images include a variety of handwritten and printed text samples, closely resembling real-life inputs such as scanned forms, notes, and informal writing.
- **Preprocessing:** Raw images are refined to improve quality and prepare them for analysis by the model. In this step, the images are resized to 32×32 pixels, and techniques such as histogram equalization are applied to enhance contrast. Binarization is also used to clearly separate text from the background. Edge

detection filters are also used to highlight the structure of the characters, making it easier for the CNN to detect meaningful patterns.

- Feature Extraction: The pre-processed images are passed through the CNN model. Unlike traditional models that rely on manually crafted features, the CNN automatically learns features such as edges, curves, and shapes from the input images using convolutional layers. Activation functions like ReLU are applied to introduce non-linearity, and max pooling layers help reduce the size of the data while preserving important information.
- Training and Optimization: This phase involves teaching the model to recognize patterns by using labeled data. Categorical cross-entropy is used as the loss function because the task involves multiple character classes. The Adam optimizer is used to update the model's weights efficiently, ensuring faster and more stable learning. Techniques such as batch normalization and dropout are added to improve training speed and reduce the risk of overfitting.
- Evaluation: Once training is complete, the Evaluation stage begins. The model's performance is measured using key metrics such as accuracy, precision, recall, and F1-score. A confusion matrix is also generated to understand where the model makes errors. Additionally, graphs showing training and validation loss and accuracy are analyzed to check if the model has learned effectively and is not overfitting.
- Prediction: The trained model is used to recognize characters in new, unseen images. The predicted text is compiled into readable strings and displayed or stored for further use. This stage demonstrates how the system can be deployed for real-world tasks like document scanning, automated data entry, or assistance for visually impaired users.

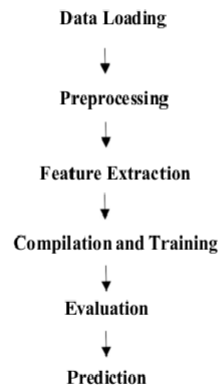


Fig 3.1: Workflow of the model

3.3 Model Architecture

The proposed text recognition system is built on a Convolutional Neural Network (CNN), chosen for its strong performance in image classification tasks, especially in identifying characters in both handwritten and printed formats. The model starts by loading labeled images from two primary sources: the EMNIST dataset and a custom dataset consisting of real-world text images captured manually. All input images are resized to a standard 32x32 pixels and converted to grayscale to maintain consistency and reduce computational complexity.

Preprocessing is then applied to enhance image clarity and ensure better recognition. This includes contrast enhancement, binarization for separating text from the background, and noise reduction to improve focus on key character features. These steps help the model better interpret variations in handwriting styles and lighting conditions.

The CNN architecture itself includes multiple layers of convolutions for extracting

features such as edges and curves, which are essential for distinguishing characters. These are followed by activation functions like ReLU to add non-linearity and max pooling layers to reduce the size of feature maps while retaining their most important aspects. The deeper layers consist of fully connected dense layers that perform classification based on the extracted features.

During training, the model uses categorical cross-entropy as the loss function and the Adam optimizer for efficient and adaptive learning. To prevent overfitting and improve generalization, techniques such as dropout and batch normalization are applied. Once trained, the model is evaluated using accuracy, confusion matrix, and loss values to ensure its performance is robust across various input types.

Finally, the trained CNN is used for predicting characters from new images, classifying them into one of 39 categories which include all English alphabets (with uppercase and lowercase combined), digits from 1 to 9, and a few special characters such as @, #, \$, and &. This architecture provides a scalable and reliable solution as shown in Fig 3.2. and also with an example as shown in Fig 3.3.

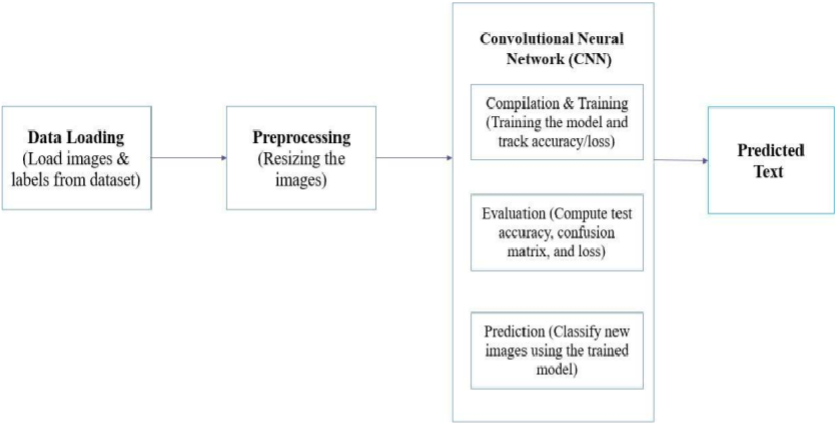
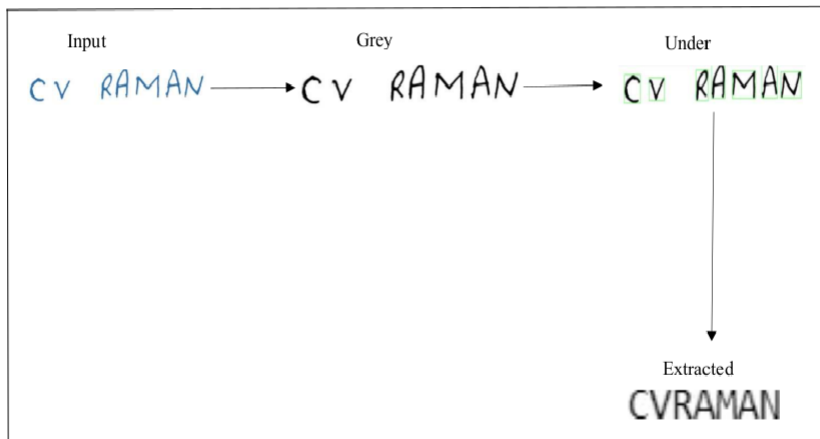


Fig 3.2: Flow Chart of Model Architecture



45
Fig 3.3: Flow Chart of the model with an example

3.4 Model Impression

The proposed CNN-based model demonstrated strong performance in recognizing both handwritten and printed text, handling a wide variety of characters with high accuracy. It was trained on a well-structured dataset that combined EMNIST data with custom images, resulting in a diverse training set that improved generalization. The model was able to distinguish between subtle variations in character shapes, thanks to its layered architecture and effective preprocessing steps.

40
During evaluation, the model showed consistent results across multiple metrics such as accuracy, loss, and confusion matrix analysis. Its ability to correctly classify characters across 39 categories, including alphabets, digits, and special symbols, indicates that it can be confidently deployed in real-world scenarios. Whether dealing with noisy, handwritten inputs or clean printed text, the model remains reliable and efficient.

Overall, the model leaves a positive impression due to its simplicity, scalability, and

ability to adapt to varying input conditions. Its performance in testing on a custom dataset captured in natural settings further validates its robustness and practical utility in OCR applications.

3.5 Deep Learning Techniques

A more advanced type of ML (Machine Learning) that uses multi-layered neural networks to automatically learn patterns, especially useful for complex tasks like image and speech recognition.

3.5.1 Cascaded Convolutional Neural Network (CNN)

Cascaded CNNs are highly effective for text detection and segmentation by leveraging hierarchical processing, where each stage refines the output of the previous one. This architecture enables the extraction of both local features, such as edges and textures, and global context, making it adept at handling various text styles, orientations, and sizes [4]. By dividing tasks into extraction, refinement, and classification phases, the CNN ensures precise focus at each stage, while multi-layer refinement reduces false positives by resolving ambiguities and consolidating overlapping text regions. This stepwise approach enhances both the accuracy and reliability of text detection in complex scenes.

The term $L_{side}(W, w)$, as shown in Eq. 1, represents the total loss computed from all side-output layers. It is primarily used to ensure that each side-output layer contributes to learning meaningful features by minimizing the discrepancy between the predicted saliency map and the ground truth. This loss allows the model to focus on both local and global information about the text regions across different layers.

$$L_{side}(W, w) = \sum_{m=1}^5 l_{side}^m(W, w^m) \quad (Eq. 1)$$

Where:

$L_{side}(W, w)$: Total loss computed from all side output.

l_{side}: Loss for the mth side output layer.

w: {w₁, w₂, ..., w₅} : A set of weights associated with different components of the model.

W: The overall model parameters involved in the fusion process.

3.5.2 Long Short-Term Memory (LSTM)

LSTMs, an advanced variant of RNNs, address the challenge of long-term dependency management by mitigating issues like vanishing gradients, making them ideal for interpreting complex, variable-length sequences. LSTMs encode referring expressions into meaningful vector representations that capture the intent and meaning of linguistic inputs, which are then fused with visual features to enhance segmentation accuracy. Their ability to adapt to variations in word order and synonyms makes LSTMs particularly useful in scenarios with diverse linguistic formulations. This flexibility ensures robust performance in real-world applications, allowing the model to handle complex text queries with precision while maintaining contextual relevance across long sequences [5].

The term $L(v_{ij}, M_{ij})$, as shown in Eq. 2, represents the weighted logistic regression loss used in the segmentation model. It ensures accurate text detection by penalizing incorrect predictions for foreground (text) and background regions. This loss balances class importance using weights α_f and α_b , addressing class imbalance and enabling precise segmentation of text regions in complex scenes.

$$L(v_{ij}, M_{ij}) = \begin{cases} \alpha_f \log(1 + \exp(-v_{ij})) & \text{if } M_{ij} = 1 \text{ (Foreground)} \\ \alpha_b \log(1 + \exp(v_{ij})) & \text{if } M_{ij} = 0 \text{ (Background)} \end{cases} \quad (Eq. 2)$$

Where:

L: The per-pixel weighted logistic regression loss.

v_{ij} : The model's predicted score for pixel (i,j).

M_{ij} : The ground truth binary label for pixel (i,j).

(1 = foreground, 0 = background).

α_f : Weight for foreground pixels.

α_b : Weight for background pixels.

Chapter 4

RESULTS AND DISCUSSIONS

4.1 Analysis of Extracted Output:

The assessment focused primarily on the performance of the system in extracting correct text information from a data set comprising images with mixed text styles, orientations, and backgrounds. The extracted output from the CNN model demonstrates high accuracy in classifying handwritten characters. The model performs well on clean, pre-processed images, with most predictions matching expected labels. The Table 4.1, 4.2 & 4.3 depict some of the important findings.

Table 4.1: Extracted output of the input images from dataset under CNN

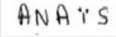
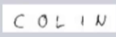
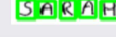
Input Image	Label	Gray Image	Extracted Output
	ADRIEN		
	ANAYS		
	COLIN		
	SARAH		
	SASHA		
	EVA		
	AMBROISE		
	AXELLE		

Table 4.2: Extracted output of handwritten text images from our custom dataset under CNN

APPLE	APPLE	APPLE	APPLE
MARKET	MARKET	MARKET	MARKET
DAILY	DAILY	DAILY	DAILY
DAY	DAY	DAY	DAY
MAGIC	MAGIC	MAGIC	MAGIC
BALL	BALL	BALL	BALL
CV RAMAN	CV RAMAN	CV RAMAN	CVRAMAN
HOLD	HOLD	HOLD	HOLD

Table 4.3: Extracted output of printed text images from our custom dataset under CNN

Input Image	Label	Gray Image	Extracted Output
	SAMPLE		
	SMILE		
	EMAIL		
	ANALYSIS		
	PEACE		

Text Detection Performance:

The CNN-based model demonstrated strong performance in detecting and classifying handwritten characters from image inputs. It effectively learned spatial features through convolutional layers, enabling accurate identification of character regions in most cases. The model performed particularly well on well pre-processed images with minimal noise. While it handled moderate distortions and handwriting variability effectively, performance slightly declined in samples with blurred strokes or overlapping characters. Overall, the CNN architecture ensured reliable character detection without the need for external segmentation or OCR components.

Model Efficiency:

The CNN model efficiently classified handwritten characters with high accuracy across varied inputs. The outputs in Table 4.2 show correct predictions for clearly written characters and occasional misclassifications in cases with distortions or overlapping strokes.

3 The model's ability to learn features from raw image data significantly enhanced performance on diverse handwriting styles without external segmentation techniques.

Challenges in Text Extraction:

- **Complex Backgrounds:** Images with busy or cluttered surroundings often led to incorrect predictions due to difficulty in distinguishing text regions.
- **Lighting Conditions:** Variations in illumination, such as shadows or glare, affected the contrast and made it hard for the CNN to isolate characters effectively.
- **Blurriness:** Some handwritten samples with overlapping strokes or low contrast caused confusion in classification.

4.2. Calculation of Evaluation Metrics:

The process of assessing a Deep learning model's performance on a particular task is known as metrics evaluation. It is an essential stage in evaluating the model's efficiency, making sure it achieves the intended goals, and pinpointing areas in need of development. Metrics serve as a scorecard while working with deep learning models, particularly for tasks like object detection or text extraction. They assist you in determining whether the model is appropriately gray regions, correctly detecting objects, or producing dependable forecasts in a range of scenarios.

Let's examine what's its significance.

- **Precision:**

7 The proportion of correctly predicted positive observations out of all predicted positive observations as shown in Eq. 3.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}} \quad (\text{Eq. 3})$$

- **Recall:**

7 The proportion of correctly predicted positive observations out of all actual positive observations as shown in Eq. 4.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}} \quad (\text{Eq. 4})$$

• **F1 Score**

A weighted average of Precision and Recall, providing a single metric that considers both false positives and false negatives as shown in Eq. 5.

$$F1\ Score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{Eq. 5}$$

• **Support**

The number of true instances for each class in the dataset (i.e., the actual ground truth distribution).

Quantitative Insights:

The model was trained and evaluated using a combination of the EMNIST dataset and a custom dataset consisting of handwritten and printed text images captured in real-world conditions. This diverse dataset provided a solid foundation for the model to learn a wide range of character variations. As shown in Table 4.4, the model achieved an accuracy score of 92.36%, indicating that it correctly predicted the character in most cases. A precision score of 93.06% shows that the model produced very few false positives, while the recall score of 92.52% reflects its ability to correctly identify actual characters. The F1-score, which balances precision and recall, was 92.64%, confirming the model's overall reliability. These results demonstrate that the CNN model is both accurate and robust, performing well on real-world text images with only minimal confusion between visually similar characters.

Table 4.4: Evaluation of Model Accuracy

Accuracy	0.92
Precision	0.93
Recall	0.92
F1 Score	0.93

4.3 Comparative Analysis

The training and validation accuracy curves in Fig 4.1. show a consistent upward trend, indicating effective learning and minimal overfitting. Both curves converge closely, suggesting the model generalizes well. Similarly, Fig 4.2. displays a steady decline in training and validation loss, confirming efficient optimization. The alignment between accuracy and loss trends reflects the model's robustness in classifying handwritten characters accurately across both seen and unseen data.

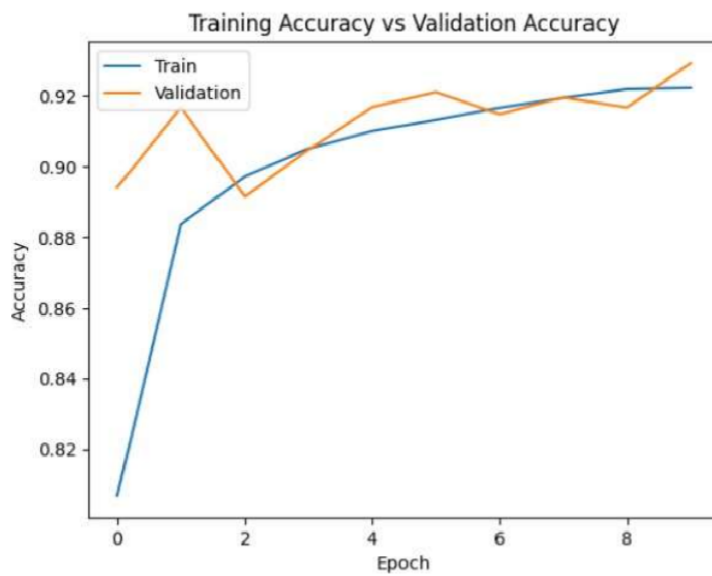


Fig 4.1: Graph between Training Accuracy and Validation Accuracy

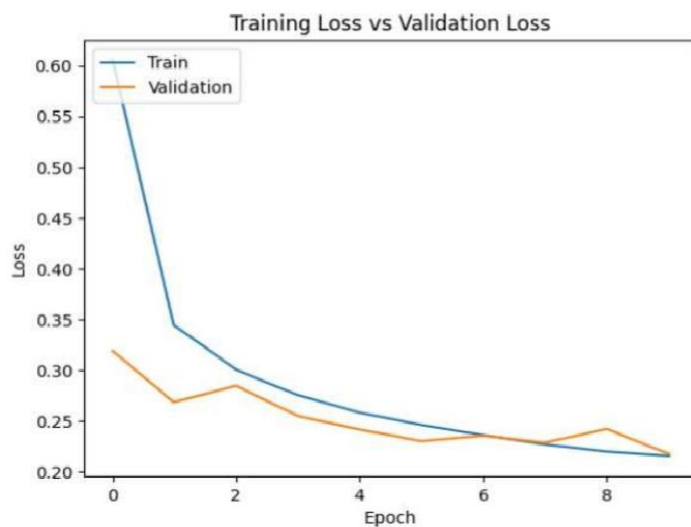


Fig 4.2: Graph between Training Loss and Validation Loss

The ROC Curve in Fig 4.3 illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) for each of the 35 character classes in the CNN model. As shown, the curves are clustered near the top-left corner, signifying excellent classification performance across all classes. This region reflects a high TPR and a low FPR, indicating that the model can accurately identify most positive instances while minimizing false alarms.

Moreover, the Area Under the Curve (AUC) values are outstanding — with the vast majority of classes achieving an AUC of 1.00 and a few reaching 0.99. These near-perfect scores confirm the model's ability to reliably distinguish between different handwritten characters. The steep rise and tight clustering of the curves further emphasize the model's robust generalization and low error rate across all classes.

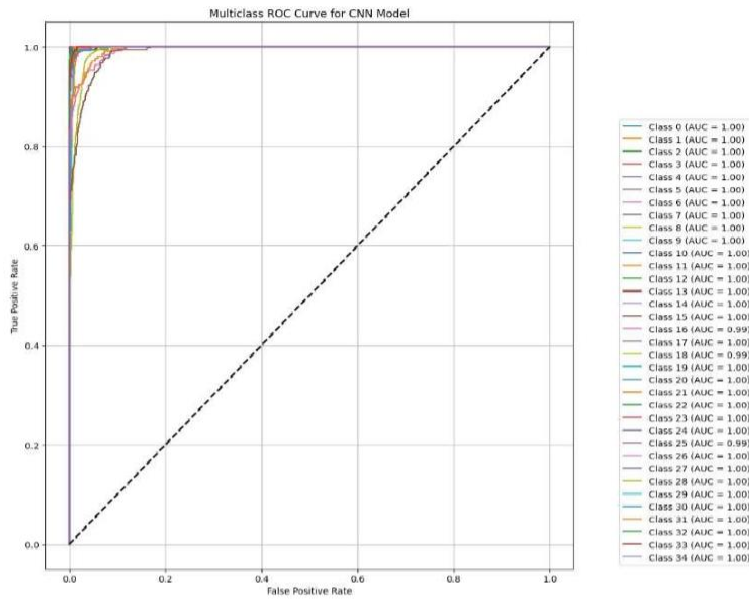


Fig 4.3: ROC of Accuracy

Challenges Identified

The outputs as shown in Table 4.2 extracted from this model despite its robustness indicate that there were minor challenges:

- **Blurring Effects:** Input images which are blurred merge together or fragment characters into partially incorrect extractions.
- **Similar Character Shapes:** Visually similar letters like 'I' and 'l' or 'O' and 'Q' occasionally caused confusion.
- **Handwritten Variability:** Inconsistent styles across samples affected uniform recognition accuracy.

CONCLUSION

This project successfully demonstrates the effectiveness of using Convolutional Neural Networks (CNNs) for text extraction from images, including both printed and handwritten text. By integrating CNNs with a structured preprocessing pipeline, the system is capable of learning complex character features directly from raw image data, eliminating the need for manual feature extraction. The model was trained on a diverse dataset composed of EMNIST samples and real-world images captured under various conditions, enabling it to generalize well across different text styles, backgrounds, and lighting scenarios.

With an achieved accuracy of 92.36%, along with high precision, recall, and F1-score values, the CNN-based model shows reliable performance in recognizing characters from noisy, distorted, and complex inputs. Its robustness and scalability make it a practical solution for real-world applications such as document digitization, automatic form processing, and assistive reading tools.

Overall, this work highlights the advantages of deep learning in OCR and paves the way for future improvements. Further enhancements could include the addition of sequence modeling for full word or sentence recognition, support for multilingual scripts, and real-time deployment on mobile or embedded devices.

In Future can be extended to this project could focus on enhancing the model's ability to recognize entire words or sentences using sequence-based models like RNNs or Transformers. Incorporating multilingual support would expand its usability across diverse regions. Real-time deployment on mobile or embedded devices can enable wider accessibility. Additionally, improving performance under noisy or distorted image conditions, adapting to various handwriting styles, and integrating Natural Language Processing (NLP) for post-recognition correction would significantly boost the system's accuracy, efficiency, and real-world applicability.

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